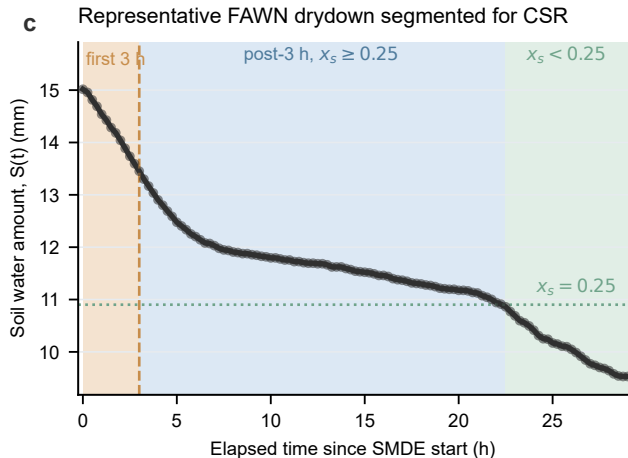
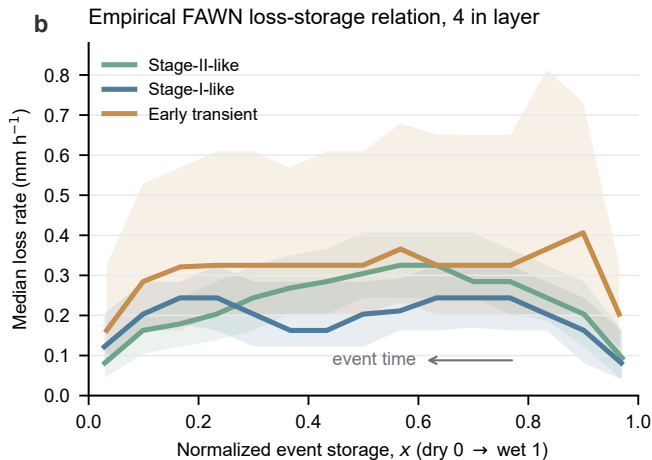
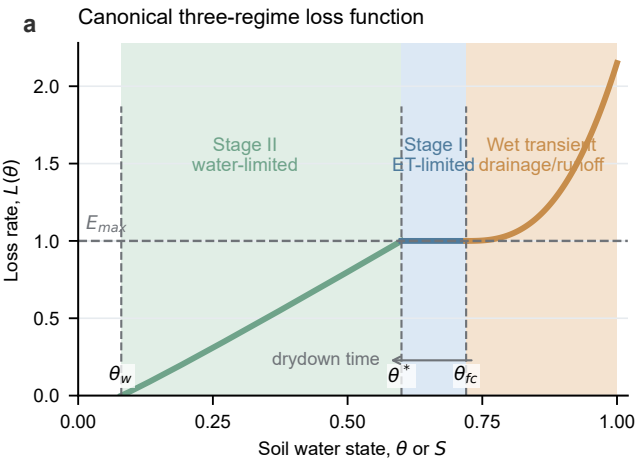


Three-regime loss-function framework used to diagnose FAWN drydowns

Conceptual loss functions define the regimes; FAWN 15-min observations provide the empirical loss-storage relation used for event filtering.



d Diagnostic equations and regime labels

Operational translation used in this study

Loss rate	$L_i = -(S_{i+1} - S_i)/\Delta t_i$
Storage coordinate	$x = (S_{mid} - S_{min})/(S_{max} - S_{min})$
Stage-II-like	$R^2_{trim3} \geq 0.70$ and $r(L, x)_{post3} \geq 0.20$
Stage-I-like	$ r(L, x)_{post3} < 0.20$ and $CV(L)_{post3} < 0.60$
Early transient-heavy	$E_3 \geq 0.40$

Labels diagnose soil-moisture behavior for event filtering; they do not partition measured fluxes.